Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A voltmeter having a resistance of 1800Ω is employed to measure the potential difference across a 200Ω resistor which is connected to the terminals of a DC power supply having an emf of $50\ \text{V}$ and an internal resistance of 20Ω. What is the percentage decrease in the potential difference across the 200Ω resistor as a result of connecting the voltmeter across it?

(A) 1%

(B) 5%

(C) 10%

(D) 25%

Q.2

The masses of the three wires of copper are in the ratio of $1:3:5$ and their lengths are in the ratio of $5:3:1$. The ratio of their electrical resistance is

(A) $1:3:5$

(B) $5:3:1$

(C) $1:25:125$

(D) $125:15:1$

Q.3

The length of a given cylindrical wire is increased by 100%. Due to the consequent decrease in diameter the change in the resistance of the wire will be

(A) 200%

(B) 100%

(C) 50%

(D) 300%

Q.4

The electric resistance of a certain wire of iron is R. If its length and radius are both doubled, then

(A) the resistance and the specific resistance, will both remain unchanged

(B) the resistance will be doubled and the specific resistance will be halved

@IITJEE_Advanced

(C) the resistance will be halved and the specific resistance will remain unchanged

(D) the resistance will be halved and the specific resistance will be doubled

Q.5

The resistance of a wire is R. It is bent at the middle by 180° and both the ends are twisted together to make a shorter wire. The resistance of the new wire is

(A) $2R$

(B) $R/2$

(C) $R/4$

(D) $R/8$

Q.6

If N, e, τ and m are representing electron density, charge, relaxation time and mass of an electron respectively, then the resistance of wire of length and cross-sectional area A is given by

(A) $\frac{\mathrm{m}\ell}{\mathrm{Ne}^2\ \mathrm{A}\tau}$

(B) $\frac{2m\tau\mathrm{A}}{\mathrm{Ne}^2\ell}$

(C) $\frac{\mathrm{Ne}^2\tau\mathrm{A}}{2\ \mathrm{m}\ell}$

(D) $\frac{\mathrm{Ne}^2\ \mathrm{A}}{2\ \mathrm{m}\tau\ell}$

Q.7

The electric resistance of a certain wire of iron is
R. If its length and radius are both doubled, then

(A) the resistance and the specific resistance, will both remain unchanged

(B) the resistance will be doubled and the specific resistance will be halved

(C) the resistance will be halved and the specific resistance will remain unchanged

(D) the resistance will be halved and the specific resistance will be doubled

Q.8

A wire X is half the diameter and half the length of a wire Y of similar material. The ratio of resistance of X to that of Y is

(A) $8:1$

(B) $4:1$

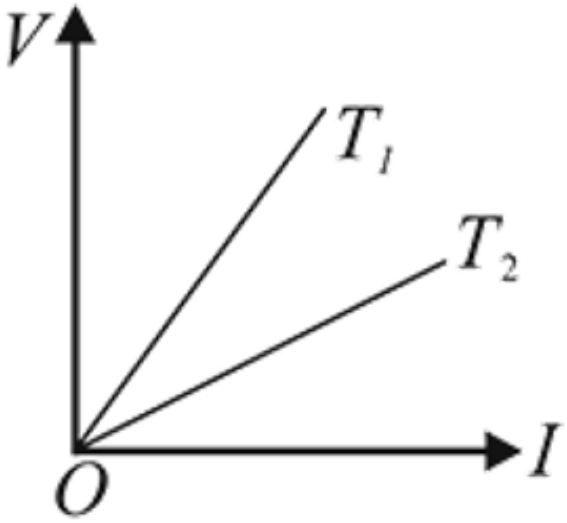
(C) $2:1$

(D) $1:1$

@IITJEE_Advanced

Q.9

The voltage V and current I graphs for a conductor at two different temperatures T_1 and T_2 are shown in the figure. The relation between T_1 and T_2 is



(A) $T_1 > T_2$

(B) $T_1 < T_2$

(C) $T_1 = T_2$

(D) $T_1 = \frac{1}{T_2}$

Q.10

The resistance of a wire is R. It is bent at the middle by 180° and both the ends are twisted together to make a shorter wire. The resistance of the new wire is

(A) $2R$

(B) $R/2$

(C) $R/4$

(D) $R/8$

Q.11

At what temperature will the resistance of a copper wire becomes three times its value at 0°C ? (Temperature coefficient of resistance of copper is $4 \times 10^{-3}/^\circ\text{C}$)

(A) 550°C

(B) 500°C

(C) 450°C

(D) 400°C

@IITJEE_Advanced

Q.12

If the resistance of a conductor is 5Ω at $50^{\circ}\mathrm{C}$& 7Ω at $100^{\circ}\mathrm{C}$, then mean temperature coefficient of resistance (of material) is

(A) $0.013/^{\circ}\mathrm{C}$

(B) $0.004/^{\circ}\mathrm{C}$

(C) $0.006/^{\circ}\mathrm{C}$

(D) $0.008/^{\circ}\mathrm{C}$

Q.13

If negligibly small current is passed though wire of length $15\ \mathrm{m}$ and resistance of 5Ω, having uniform cross section of $6 \times 10^{-7}\ \mathrm{m}^2$, then coefficient of resistivity of material is

(A) $1 \times 10^{-7}\Omega - \mathrm{m}$

(B) $2 \times 10^{-7}\Omega - \mathrm{m}$

(C) $3 \times 10^{-7}\Omega - \mathrm{m}$

(D) $4 \times 10^{-7}\Omega - \mathrm{m}$

Q.14

The resistance of a wire at room temperature $30^{\circ}\mathrm{C}$ is found to be 10Ω. Now to increase the resistance by 10%, the temperature of the wire must be [The temperature coefficient of resistance of the material of the wire is 0.002 per $^{\circ}\mathrm{C}$]

(A) $36^{\circ}\mathrm{C}$

(B) $83^{\circ}\mathrm{C}$

(C) $63^{\circ}\mathrm{C}$

(D) $33^{\circ}\mathrm{C}$

Q.15

A wire of a certain material is stretched slowly by ten per cent. Its new resistance and specific resistance become respectively:

(A) 1.2 times, 1.3 times

(B) 1.21 times, same

(C) both remain the same

(D) 1.1 times, 1.1 times

@IITJEE_Advanced

Q.16

A 4 ohm resistance wire is bent through 180° at its mid point and the two halves are twisted together. Then the resistance is

(A) 1Ω

(B) 2Ω

(C) 5Ω

(D) 8Ω

Q.17

The length of a given cylindrical wire is increased by 100%. Due to the consequent decrease in diameter the change in the resistance of the wire will be

(A) 200%

(B) 100%

(C) 50%

(D) 300%

Q.18

A uniform wire of length $\ell = 4m$ and radius r has a resistance of 100Ω. It is recast into a wire of radius $\frac{\mathrm{r}}{2}$. The resistance of new wire will be :

(A) 1600Ω

(B) 400Ω

(C) 200Ω

(D) 100Ω

Q.19

A wire has a resistance 12Ω. It is bent in the form of circle. The effective resistance between two points on any diameter is

(A) 6Ω

(B) 3Ω

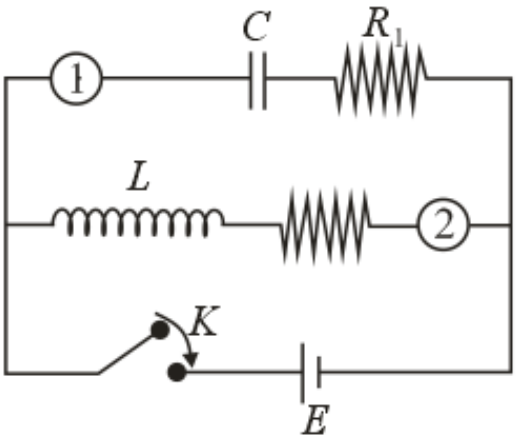
(C) 12Ω

(D) 24Ω

Q.20

@IITJEE_Advanced

In the circuit given below 1 and 2 are ammeters. Just after key K is pressed to complete the circuit, the reading is :



(A) Maximum in both 1 and 2

(B) Zero in both 1 and 2

(C) Zero in 1, minimum in 2

(D) Maximum in 1 , zero in 2

Answers & Solutions

Q.1 Answer:

1%

Solution:

Current $I = \frac{50}{(200+20)} = \frac{5}{22}$ A

$\therefore$ Potential drop across 200Ω resistor

$(V) = \frac{5}{22} \times 200 = \frac{500}{11}$ V

When a voltmeter of resistance 1800Ω is connected across the 200Ω resistor, the effective resistance R is given by

$\frac{1}{R} = \frac{1}{1800} + \frac{1}{200}$ which gives $R = 180\Omega$

The current in the circuit becomes

$I' = \frac{50}{180+20} = \frac{5}{20}$ A

The potential drop becomes

$V' = \frac{5}{20} \times 180 = 45$ V

Difference $V - V' = \frac{500}{11} - 45 = \frac{5}{11}$ V

Percentage decrease $= \frac{5}{11} \times \frac{11}{500} \times 100 = 1\%.$

Hence the correct choice is (a).

Q.2 Answer:

$125 : 15 : 1$

Solution:

@IITJEE_Advanced

$R = \frac{\rho l}{\pi r^2}$. **But** $m = \pi r^2 l d \therefore \pi r^2 = \frac{m}{ld}$
$\therefore R = \frac{\rho l^2 d}{m}, R_1 = \frac{\rho l_1^2 d}{m_1}, R_2 = \frac{\rho l_2^2 d}{m_2}$
$R_3 = \frac{\rho l_3^2 d}{m_3}$
$R_1 : R_2 : R_3 = \frac{l_1^2}{m_1} : \frac{l_2^2}{m_2} : \frac{l_3^2}{m_3}$
$R_1 : R_2 : R_3 = \frac{25}{1} : \frac{9}{3} : \frac{1}{5} = 125 : 15 : 1$

Q.3 Answer:

300%

Solution:

The total volume remains the same before and after stretching.
Therefore $A \times \ell = A' \times \ell'$
Here $\ell' = 2\ell$
$\therefore A' = \frac{A \times \ell}{\ell'} = \frac{A \times \ell}{2\ell} = \frac{A}{2}$
Percentage change in resistance
$= \frac{R_f - R_i}{R_i} \times 100 = \frac{\rho\left(\frac{\ell'}{A'} - \frac{\ell}{A}\right)}{\rho\frac{\ell}{A}} \times 100$
$= \left[\left(\frac{\ell'}{A'} \times \frac{A}{\ell}\right) - 1\right] \times 100 = \left[\left(\frac{2\ell}{A/2} \times \frac{A}{\ell}\right) - 1\right] \times 100$
$= 300\%$

Q.4 Answer:

the resistance will be halved and the specific resistance will remain unchanged

Solution:

$R = \frac{\rho \ell_1}{A_1}$, **now** $\ell_2 = 2\ell_1$
$A_2 = \pi (r_2)^2 = \pi (2r_1)^2 = 4\pi r_1^2 = 4A_1$
$\therefore \quad R_2 = \frac{\rho(2\ell_1)}{4A_1} = \frac{\rho \ell_1}{2A_1} = \frac{R}{2}$
$\therefore$ **Resistance is halved, but specific resistance remains the same.**

Q.5 Answer:

$R/4$

Solution:

Resistance of wire $(R) = \rho\frac{l}{A}$ **If wire is bent in the middle then** $l' = \frac{l}{2}, A' = 2A$
$\therefore \quad$ **New resistance,** $\mathrm{R}' = \rho\frac{l'}{\mathrm{A}'} = \frac{\rho\frac{l}{2}}{2A}$
$= \frac{\rho l}{4A} = \frac{R}{4}$

Q.6 Answer:

$\frac{\mathrm{m}\ell}{\mathrm{Ne}^2\ \mathrm{A}\tau}$

Solution:

@IITJEE_Advanced

No solution available

Q.7 Answer:

the resistance will be halved and the specific resistance will remain unchanged

Solution:

$\mathrm{R}=\frac{\rho \ell_1}{\mathrm{A}_1}$, **now** $\ell_2=2 \ell_1$

$\mathrm{A}_2=\pi\left(\mathrm{r}_2\right)^2=\pi\left(2 \mathrm{r}_1\right)^2=4 \pi \mathrm{r}_1^2=4\ \mathrm{A}_1$

$\mathrm{R}_2=\frac{\rho\left(2 \ell_1\right)}{4\ \mathrm{A}_1}=\frac{\rho \ell}{2\ \mathrm{A}}=\frac{\mathrm{R}}{2}$

Resistance is halved, but specific resistance remains the same

Q.8 Answer:

$2:1$

Solution:

$\mathrm{R}=\frac{\rho \ell}{\left(\pi \mathrm{D}^2 / 4\right)}$ **or** $\mathrm{R} \propto \frac{\ell}{\mathrm{D}^2}$

$\frac{\mathrm{R}_{\mathrm{x}}}{\mathrm{R}_{\mathrm{y}}}=\frac{\ell_{\mathrm{x}}}{\mathrm{D}_{\mathrm{x}}^2} \times \frac{\mathrm{D}_{\mathrm{y}}^2}{\ell_{\mathrm{y}}}=\frac{\ell_{\mathrm{y}} / 2}{\left(\mathrm{D}_{\mathrm{y}} / 2\right)^2} \times \frac{\mathrm{D}_{\mathrm{y}}^2}{\ell_{\mathrm{y}}}=\frac{2}{1}$

Q.9 Answer:

$T_1>T_2$

Solution:

The slope of V-I graph gives the resistance of a conductor at a given temperature. From the graph, it follows that resistance of a conductor at temperature T_1 is greater than at temperature T_2. As the resistance of a conductor is more at high temperature and less at lower temperature, hence $\mathrm{T}_1>\mathrm{T}_2$.

Q.10 Answer:

$R/4$

Solution:

Resistance of wire $(R)=\rho \frac{l}{A}$ **If wire is bent in the middle then** $l^{\prime}=\frac{l}{2}, A^{\prime}=2 A$

$\therefore$ **New,** $\mathrm{R}^{\prime}=\rho \frac{l^{\prime}}{\mathrm{A}^{\prime}}=\frac{\rho \frac{l}{2}}{2 A}=\frac{\rho l}{4 A}=\frac{R}{4}$.

Q.11 Answer:

$500^{\circ} \mathrm{C}$

Solution:

$\mathrm{R}_{\mathrm{t}}=\mathrm{R}_0(1+\alpha \mathrm{t})$ **at** $\mathrm{t}^{\circ} \mathrm{C} \mathrm{R}_{\mathrm{t}}=3 \mathrm{R}_0$

$\alpha=4 \times 10^{-3} /^{\circ} \mathrm{C}$

$3 \mathrm{R}_0=\mathrm{R}_0\left(1+4 \times 10^{-3} \times \mathrm{t}\right)$

@IITJEE_Advanced

$\therefore \quad 3-1=4\times10^{-3}\mathrm{t}$

$\therefore \quad \mathrm{t}=\frac{2}{4\times10^{-3}}=500^{\circ}\mathrm{C}$

Q.12 Answer:

$0.013/^{\circ}\mathrm{C}$

Solution:

[**Hint** $\Rightarrow R_t=R_0(1+\alpha t)$]

$5\Omega=\mathrm{R}_0(1+\alpha\times50)$ **and** $7\Omega=\mathrm{R}_0(1+\alpha\times100)$

or $\frac{5}{7}=\frac{1+50\alpha}{1+100\alpha}$ **or** $\alpha=\frac{2}{150}=0.0133/^{\circ}\mathrm{C}$

Q.13 Answer:

$2\times10^{-7}\Omega-\mathrm{m}$

Solution:

Hint $\Rightarrow \rho=\frac{\mathrm{R.A}}{\ell}$ **= Coefficient of resistivity)**

Q.14 Answer:

$83^{\circ}\mathrm{C}$

Solution:

Given: Temperature coefficient of resistance $=\alpha=0.002/^{\circ}\mathrm{C}$

We know that, $\mathrm{R_T}=\mathrm{R}_0(1+\alpha\mathrm{T})$

Initially, $10=\mathrm{R}_0(1+\alpha.30)\ldots(1)$

After 10% **increase the resistance becomes** $=10+10\times\frac{10}{100}=11\Omega$ **and corresponding temp is T.**

Now, $11=\mathrm{R}_0(1+\alpha\mathrm{T})\ldots(2)$

$(2)/(1)\Rightarrow\frac{11}{10}=\frac{\mathrm{R}_0(1+\alpha\mathrm{T})}{\mathrm{R}_0(1+30\alpha)}$

$1+\alpha\mathrm{T}=1.1+33\alpha$

$\mathrm{T}=\frac{1.1+33\alpha-1}{\alpha}=\frac{0.1+33\times0.002}{0.002}=83^{\circ}\mathrm{C}$

Q.15 Answer:

1.21 times, same

Solution:

$$\mathrm{R}=\frac{\rho l}{\mathrm{A}}$$

Now, $1=1+\frac{1}{10}=\frac{111}{10}$

and therefore, $A=\frac{10\ \mathrm{A}}{11}$

So $\mathrm{R}'=\frac{\mathrm{p}\times\left(\frac{111}{10}\right)}{\frac{10\ \mathrm{A}}{11}}=\frac{\rho l}{\mathrm{A}}\times\frac{(11)^2}{(10)^2}=1.21\mathrm{R}$

@IITJEE_Advanced

Now resistance becomes 1.21 times of initial and specific resistance is the intrinsic property so remains same.

Q.16 Answer:

1Ω

Solution:

$\mathrm{R}=\rho \ell / \mathrm{A} ; \mathrm{R}^{\prime}=\frac{\rho \ell^{\prime}}{\mathrm{A}^{\prime}}=\frac{\rho(\ell / 2)}{2 \mathrm{~A}}=\frac{1}{4} \mathrm{R}.$

Since here $\mathrm{R}=4\Omega$

So resistance $=4 \times(1 / 4) \Omega=1 \Omega$

Q.17 Answer:

300%

Solution:

Here the total volume of the wire is constant.

Resistance $\mathrm{R}=\frac{\rho \mathrm{l}^2}{\mathrm{V}}$ **or** $\mathrm{R} \propto \mathrm{l}^2$

let, initial length $l_1=1_0$ **and final length** $l_2=1_0+1_0 \times \frac{100}{100}=21_0$ **thus,** $\frac{\mathrm{R}_2}{\mathrm{R}_1}=\frac{1_2^2}{1_1^2}=\frac{41_0^2}{1_0^2}=4$

The change in resistance $=\frac{\mathrm{R}_2-\mathrm{R}_1}{\mathrm{R}_1} \times 100=\left[\frac{\mathrm{R}_2}{\mathrm{R}_1}-1\right] 100=(4-1) 100=300$ **96**

Q.18 Answer:

1600Ω

Solution:

Uniform wire length 1

Redius r

Resistance 100Ω

If its radius $=\frac{r}{2}$

$\Rightarrow \frac{\mathrm{r}_2}{\mathrm{r}_1}=\frac{\frac{\mathrm{l}_2}{\mathrm{A}_2}}{\frac{\mathrm{l}_1}{\mathrm{A}_1}}$

volume remians constant

$$\Rightarrow 1 \times \pi \mathrm{r}^2=1 \times \pi \frac{\mathrm{r}^2}{4}$$

We know, $l=4]$

$$\Rightarrow \mathrm{r}_2=\mathrm{r}_1\left[\frac{4}{\pi \mathrm{r}^2 / 4} \frac{1}{\pi \mathrm{r}^2}\right]$$

$$\Rightarrow \mathrm{r}_2=16 \mathrm{r}_1$$

@IITJEE_Advanced

$$\Rightarrow r_2 = 1600\pi$$

Q.19 Answer:

3Ω

Solution:

Resistance of the wire of a semicircle $= 12/2 = 6\Omega$

For equivalent resistance between two points on any diameter, 6Ω **and** 6Ω **are in parallel**

$\frac{1}{6} + \frac{1}{6} = \frac{6}{2} = 3.$

Q.20 Answer:

Maximum in 1 , zero in 2

Solution:

Initially capacitor provides no resistance to current and hence maximum current flows through 1. Moreover initially Inductor provides infinite resistance to current and hence zero current flows through 2.

@IITJEE_Advanced